Spend Categorization

Databricks AI Accelerators

January 7, 2025

Executive Summary

Spend categorizaiton can be solved internally

Better spend categorization could save millions via procurement optimization

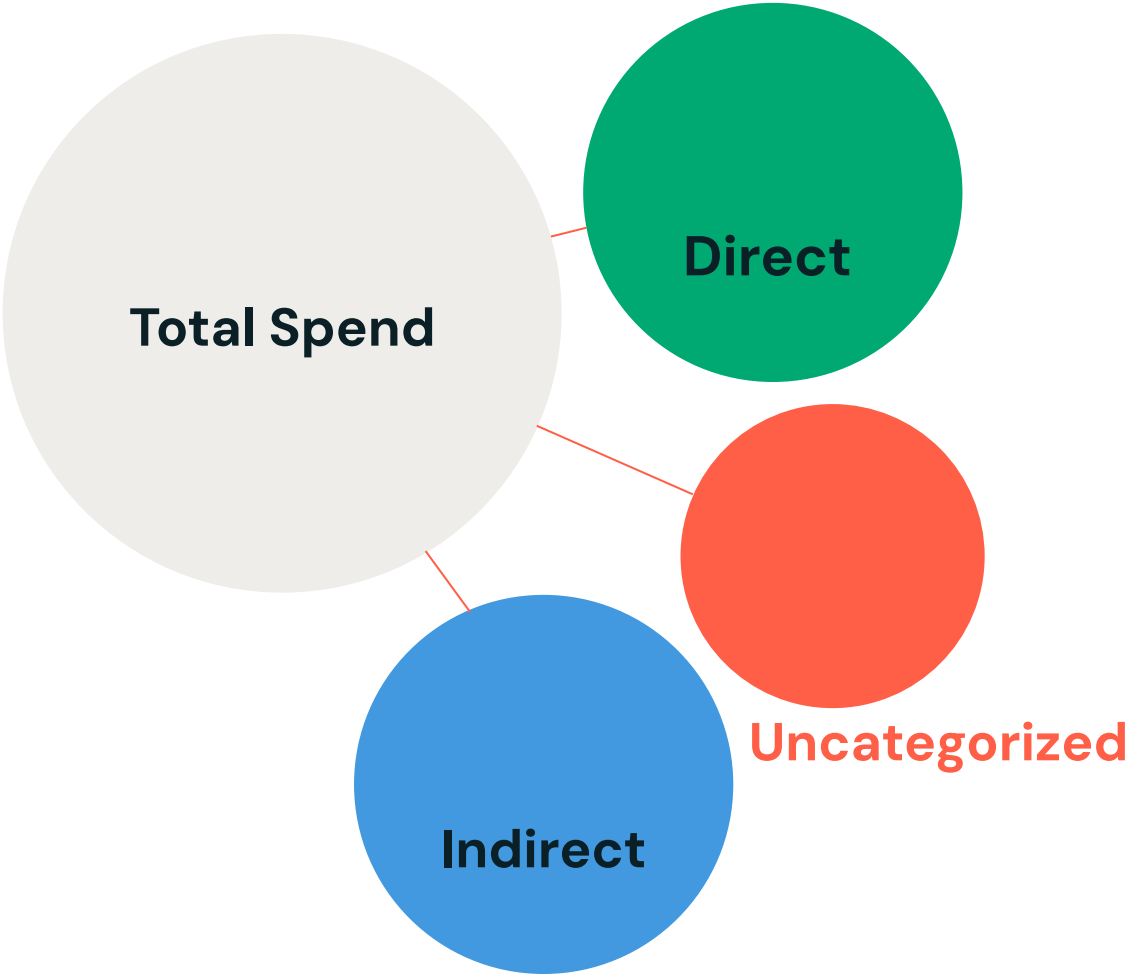
Rule-based solutions have degrading performance and high maintenance

We show two solutions with high accuracy and low maintenance:
retrieval augmented generation and tree-based machine learning

Both approaches are maintained by correcting poor categorization at the source

Problem Summary

A lot of spend is uncategorized



Most existing solutions are rules-based and need a lot of maintenance.

Key Questions

Can we automate the solution?

Can it beat the existing system?

How much will it cost?

Solution Overview

One solution from ERP to UI

ERP Systems

Classification Hierarchy

Unity Catalog Tables

Traditional ML

Catboost Model

Sequential Prediction

Vector Search

Foundation Model

All Level Prediction

Generative AI

Correct Raw Data

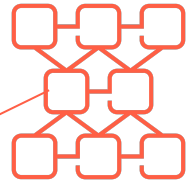
Category Review Interface

Award Dataset

Awards & transaction from US government spending

USASPENDING.gov

We are using a public database of federal awards as an analogy to a spend categorization dataset.



Manually classified award agencies are in the transactions table

Award Hierarchy

The agency and sub agency are analogous to spend categories. The hierarchy can be managed separately from the transactions.

Transactions

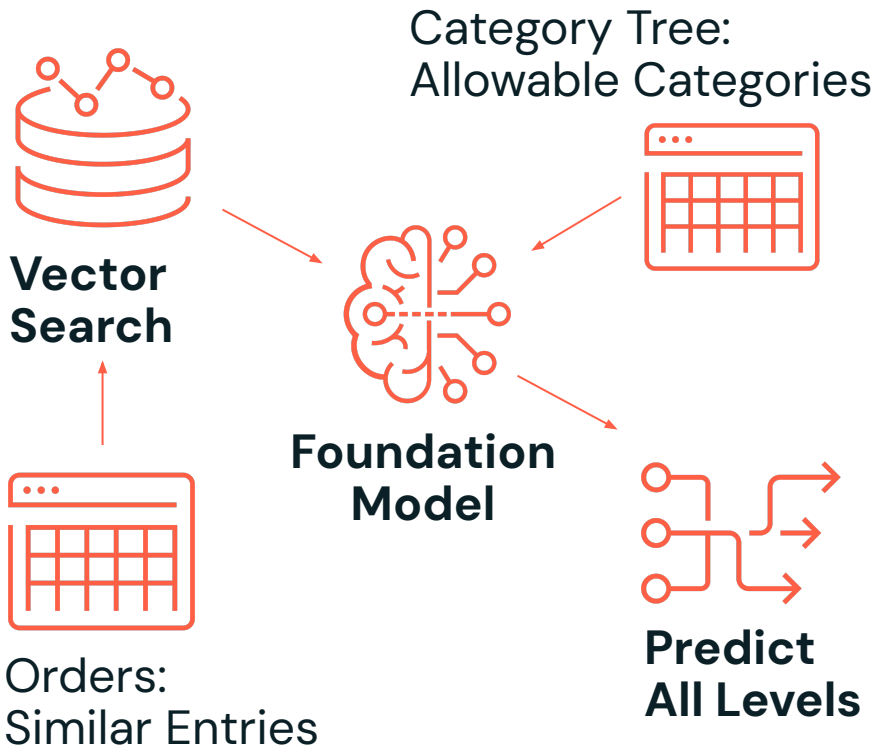
Contains 'features', which are almost all text and our 'targets' – the agency classifications from manual entries or an existing rule-based system.

Solution 1: Retrieval Augment Generation

Engineer context

Naive LLM: 53% accuracy

RAG: 96% accuracy*

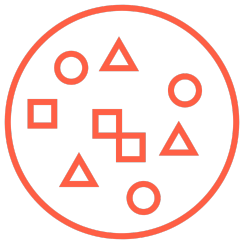


	Precision	Recall	Spend
Department of Defense	1	1	17884
Department of Homeland Security	**0.94**	1	5271
Department of Health and Human Services	1	**0.98**	2291
Department of Justice	1	1	1909
Environmental Protection Agency	**0.97**	1	1578
Department of the Interior	**0.99**	1	1330
Securities and Exchange Commission	1	1	1160
International Development Finance	1	1	581
Department of State	**0.91**	**0.88**	529
Department of Commerce	1	1	329
Department of Veterans Affairs	0	0	288
Smithsonian Institution	1	**0.93**	241

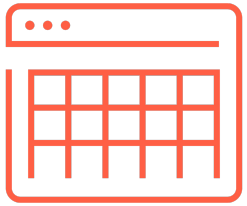
*98% of spend dollars is correctly categorized

Solution 1: Retrieval Augment Generation

When querying LLMs, we use structured outputs to enforce hierarchy / ontology
We could also transition to tools / agent framework



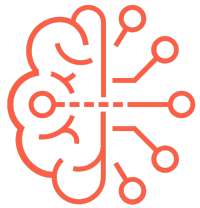
Unstructured Data → Structured Data



user: morgan.smith@gmail.com
query: Hi, I'm a welder than needs help with my equipment, can you direct me to maintenance?

Query

Language Model



name: "morgan"
occupation: "welder"
topic: "maintenance"

Structured Response

Schema

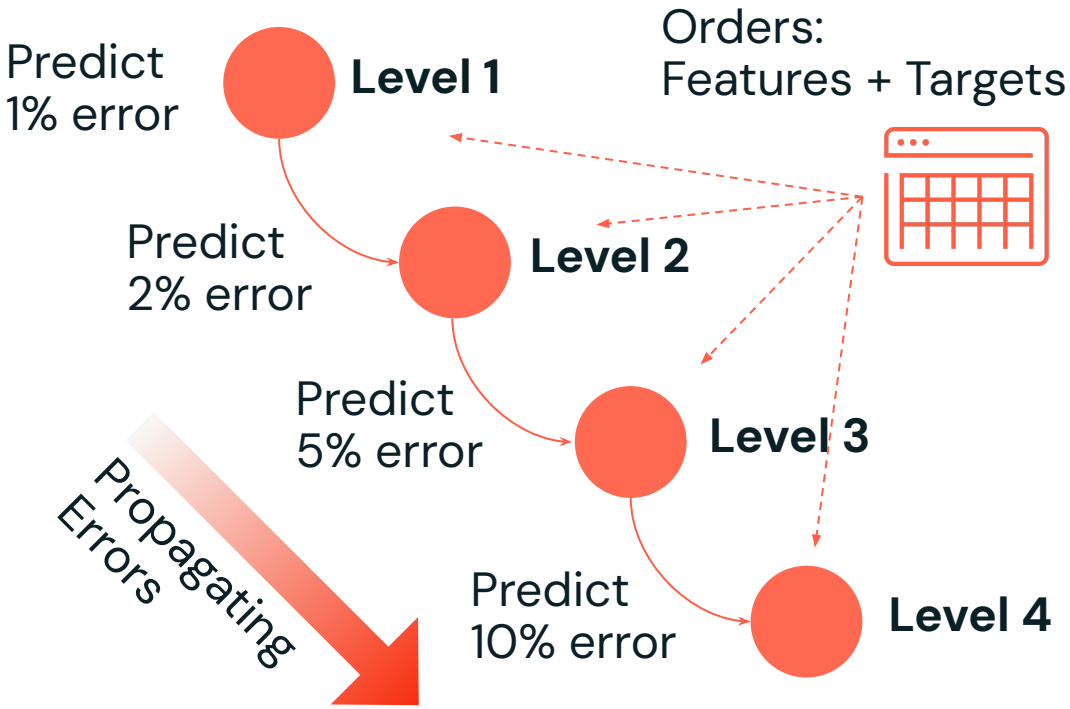
name: string
occupation: string
topic: string

Solution 2: Machine Learning

Multilevel modeling

Automates existing solution

94% accuracy



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Department of the Interior	**0.85**	1	1330
Securities and Exchange Commission	1	1	1160
International Development Finance	1	**0.73**	581
Department of State	**0.68**	**0.75**	529
Department of Commerce	1	1	329
Department of Veterans Affairs	1	**0.97**	288
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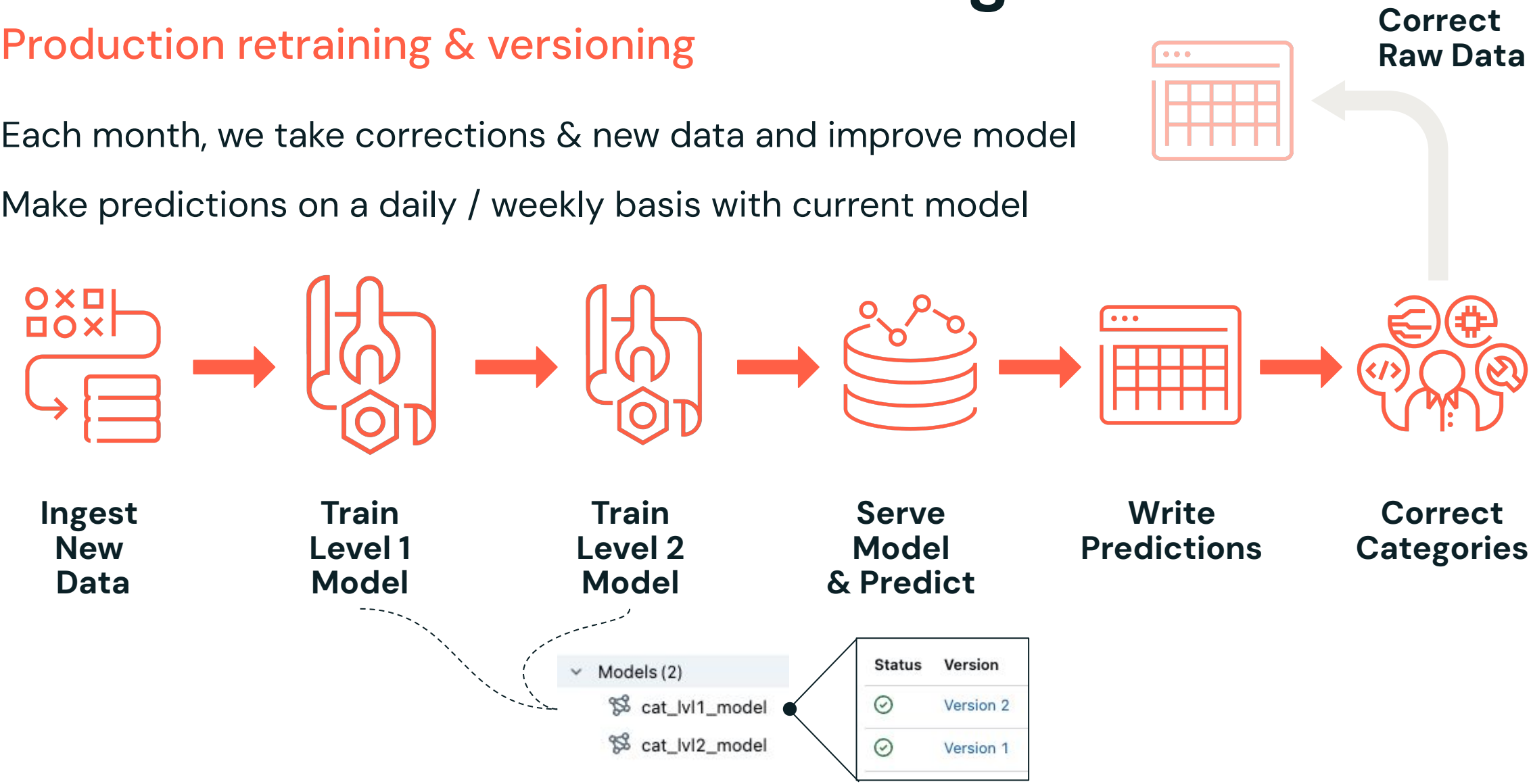
***Tree models naturally enforce category ontology, but 100% match verified

Solution 2: Machine Learning

Production retraining & versioning

Each month, we take corrections & new data and improve model

Make predictions on a daily / weekly basis with current model